

BV-VPIN: Measuring the impact of order flow toxicity and liquidity on international equities markets[☆]

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Abstract

Order flow toxicity is a measure of a trader's exposure to the risk that counter-parties possess private information or other informational advantages. High levels of order flow toxicity can culminate in market makers providing liquidity at a loss or in suboptimal execution of trades. From a regulatory perspective, high levels of toxicity can be harmful to overall market liquidity and precede precipitous drops in asset prices. Bulk Volume VPIN (BV-VPIN) is one way of measuring the "toxicity" component of order flow that has been successfully applied in High Frequency Trading (HFT) environments. We apply the BV-VPIN to daily data for a range of international indices to extend previous analysis of its properties. We find that a rise in BV-VPIN effectively foreshadows high-levels of volatility in the equities indices of several countries. If a BV-VPIN futures contract exists, we show that it would exhibit safe haven characteristics during market downturns. In particular, a simple active portfolio management strategy that times investments in equities (risk-free asset) when BV-VPIN levels are low (high) outperforms a buy-and-hold strategy. Thus, we find support for the application of BV-VPIN in international equities markets as a risk monitoring and management tool for portfolio managers and regulators.

Keywords: VPIN, Bulk Volume Classification, international equities, liquidity, order flow toxicity

JEL classification: G12, G14, C58, D53

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1. Introduction

On May 6th 2010, the Dow Jones Industrial Average (DJIA) suffered a decline of 998.5 points. This event is the largest one-day point decline of the DJIA, and regularly referred to as the “flash crash”. Several explanations have been provided, e.g., speculative trading by hedge funds (Patterson and Lauricella, 2010; Phillips, 2010), currency movements in the US dollar/Japanese yen exchange rate (Krasting, 2010), and technical reporting difficulties on the NYSE (Flood, 2010). However, the most salient feature of the flash crash is the extreme lack of liquidity that occurred (CFTC-SEC, 2010a,b; Easley et al., 2011b). More specifically, Easley et al. (2011b) attribute that tight liquidity to an unprecedented increase in order flow toxicity, where uninformed traders (e.g., market makers) start to reduce their positions significantly to curtail risks of significant losses due to adverse selection by informed traders (e.g., hedge funds). Such actions are accompanied by market illiquidity and a significant fall in asset prices.

Easley et al. (2012) develop the Volume-Synchronized Probability of Informed Trading (VPIN) model as a measure of order flow toxicity and Easley et al. (2011b) show that the cumulative distribution function (CDF) of VPIN would have been able to predict the “flash crash” more than an hour in advance. The VPIN is a metric of the probability that the liquidity provision process might fail, thus resulting in adverse price movements. Our work investigates the application of a version of VPIN, viz. Bulk Volume VPIN (BV-VPIN) to daily international equities data to evaluate its effectiveness in ascertaining the probability of sharp market movements, and its ability to act as a safe haven or hedging tool if implemented via a futures contract.

Comparing a range of trade classification algorithms that use tick data and the bulk volume of trades, Easley et al. (2016) find that BV-VPIN is closely linked to information-based trading proxies such as Hi-Lo spreads and the permanent price effect of trades. BV-VPIN appears incrementally better than the original VPIN in discerning trade motivation from market data. Internationally, Abad and Yage (2012) apply both BV-VPIN and Probability of Informed Trading (PIN) on the Spanish equities market and conclude that BV-VPIN is a useful proxy for adverse selection risk. They find in particular that the Volume Bucket Size (VBS) input of the BV-VPIN can be adjusted to capture transitory or permanent information in the trade data being analyzed, and further conclude that BV-VPIN has potential applications in the context of low-frequency analyses, and

should not be limited to purely High Frequency Trading (HFT) environments.¹

The BV-VPIN methodology has also been the subject of criticism. Andersen and Bondarenko (2014b) find contradictory results to those of Easley et al. (2011b) and document that BV-VPIN is a poor predictor of short-term volatility. They suggest that any predictive content is due to a mechanical relation with underlying trading intensity. However, Easley et al. (2014) argue that the findings of Andersen and Bondarenko (2014b) are due to confusion in the methodology, analysis and conclusions drawn from their work. Furthermore, Wu et al. (2013) analyze the 94 most active futures contracts and find BV-VPIN to be a strong predictor of liquidity-induced volatility. Andersen and Bondarenko (2014a) and Andersen and Bondarenko (2015) continue to report that transaction-based classification schemes are more accurate than bulk volume strategies advocated by Easley et al. (2011b) and Easley et al. (2012).

Given the apparent importance of implementation and market context to the predictive content of BV-VPIN as an early warning signal of impending volatility, it is instructive to apply it across a variety of international equity markets using daily data. That is the main purpose of our paper. We analyze the impact of different values for the VBS and the sample size applied as inputs into BV-VPIN model based on the US market, to ascertain the optimal criteria for application across all other countries in our dataset. We report the threshold value that is optimal across a variety of countries. The economic value of BV-VPIN is evaluated by applying a scenario where a fund manager exhibits a flight-to-quality action by alternating investment between a riskless security and the stock market when the BV-VPIN is above or below the threshold value over the length of our dataset. Our study includes the evaluation of the safe haven and hedge properties of BV-VPIN in the context of international equities markets.

We also show that BV-VPIN has potential application beyond that of HFT. The CDF of BV-VPIN shows promise as a long-term predictor of market volatility. BV-VPIN also exhibits safe haven and hedge characteristics during well-known crisis periods, e.g., Asian Financial Crisis (AFC), Dot-com Bubble (DCB), Great Recession (GR), and the US Credit Rating Downgrade (USCRD). Incorporating BV-VPIN as a flight-to-quality indicator in an asset management application results in substantial out performance of equity benchmark strategies for four out of six countries analyzed in our study.

¹ Given regulatory interests in promoting efficient market trading mechanisms, Easley et al. (2011b) and Bell (2013) recommend that regulators use BV-VPIN as an early warning signal to herald the implementation of regulatory action to forestall crashes or identify unusual market conditions. Easley et al. (2012) shows that BV-VPIN has forecasting power over volatility (toxicity-induced) and advocate its use as a valuable risk management tool for market making activity. BV-VPIN has application for trading strategies based on volatility arbitrage and for brokers who seek to ascertain the optimal time of execution (Easley et al., 2015). Easley et al. (2011a) detail the specifications of a BV-VPIN contract that could be used as a hedge against the risk of higher than expected levels of toxicity and to monitor such risk. VPIN has been applied in the empirical finance literature to evaluate the market reaction to public and private information (Vega, 2006), market anomalies (Kang, 2010; Chen and Zhao, 2012), asset pricing (Aslan et al., 2011), and more. Corcoran (2012) discuss how VPIN can be applied to detect mini-bubbles based upon the idea of monitoring the probability of toxicity-induced liquidity crises.

Our contributions are threefold. First, we apply the BV-VPIN to an international equities dataset including US, UK, Germany, Japan, Australia, and China and analyze its performance over a dataset spanning 16 years. To the best of our knowledge, no other studies have carried out such a comprehensive analysis that applies VPIN-based measures across different international markets. Second, our dataset allows us to determine the optimal thresholds for VPIN to predict large down movements across a range of international equities markets. Third, we analyze whether BV-VPIN is useful for safe haven and hedging properties if it were implemented as a futures contract as suggested by Easley et al. (2011a).

The remainder of this paper is organized as follows. Section 2 presents the descriptive statistics of our datasets and Section 3 details the BV-VPIN algorithm and parameters selected in our study. The results and conclusions of our study are reported in Section 4 and Section 5, respectively.

2. Data

Our dataset includes daily equity returns and volume data for US (S&P500), UK (FTSE100), German (DAX), French (CAC40), Japanese (Nikkei 225), Chinese (SHCOMP), and Australian (AS30) indices, sourced from the Bloomberg data terminal. The period covered is 1995 to 2015 for all countries except China. For China, our dataset extends from 1997 to 2015.

Table 2 provides descriptive statistics for our data of both daily returns and volume data for all indices. Japan is the only country to have a negative mean return. The standard deviation of daily returns is similar across all countries, with Australia exhibiting the lowest. All indices exhibit negative skewness. The US exhibits the highest levels of kurtosis, and Germany and China the lowest. The null hypotheses of normality is rejected at the 1% for all countries using the Jarque-Bera test.² The daily volume characteristics are such that most countries exhibit similar mean daily volumes with China (Germany) being the highest (lowest). The daily volume data of China also exhibits the highest positive skewness and kurtosis.

3. Measures of order flow toxicity

VPIN is related to order flow toxicity. The higher the order flow toxicity, the greater the potential for a negative impact for market makers from being adversely selected by informed traders. When toxicity levels are too high, market makers will leave the market resulting in a reduction in liquidity and short-term toxicity-induced volatility.

VPIN uses volume imbalance to signal the toxicity of order flow, and the overall level of volume determines the frequency for VPIN metric updates. VPIN is updated based upon fixed units of volume, rather than fixed units of time. The most important feature of VPIN is its emphasis

²For brevity, we do not report the Jarque-Bera test-statistic. These results can be provided upon request.

Table 1: Descriptive statistics of daily returns and trading volume of international indices

This table shows the descriptive statistics of daily returns and volume data for indices from 6 different countries. Our dataset spans from January 1995 to December 2015, except for the SHCOMP (China) that spans from November 1997 to December 2015. The sample size indicates the number of days of data for each equities index. The mean, minimum, and maximum are presented as percentages.

Countries	Indices	Mean	Std. Dev.	Skew	Kurt.	Min.	Max.	Sample size
<i>Panel A: Daily Returns</i>								
United States	S&P 500	0.03	0.0121	-0.2449	8.0496	-9.47	10.96	5288
United Kingdom	FTSE 100	0.01	0.0117	-0.1639	5.9236	-9.27	9.38	5334
Germany	DAX	0.03	0.0150	-0.1273	4.1825	-8.87	10.80	5320
Japan	Nikkei 225	-0.0007	0.0153	-0.2901	5.3782	-12.11	13.23	5160
China	SHCOMP	0.02	0.0164	-0.2585	4.3751	-9.26	9.40	4355
Australia	AS30	0.02	0.0095	-0.5590	6.4182	-8.55	6.07	5315
<i>Panel B: Daily Volume</i>								
United States	S&P 500	9.39e+08	4.87e+08	0.5390	-0.2553	1.23e+06	2.96e+09	5288
United Kingdom	FTSE 100	9.98e+08	6.06e+08	0.6116	0.1695	5.00e+00	4.41e+09	5334
Germany	DAX	8.78e+07	6.25e+07	1.1347	3.3525	7.72e+05	4.97e+08	5320
Japan	Nikkei 225	9.48e+08	6.38e+08	1.0151	2.4677	7.09e+07	5.95e+09	5160
China	SHCOMP	7.75e+09	1.10e+10	3.1178	11.9300	1.23e+08	8.57e+10	4355
Australia	AS30	6.33e+08	4.34e+08	1.0159	3.5692	3.48e+05	5.64e+09	5315

on volume as a critical variable in understanding price adjustments and its linkage to underlying information dynamics. In HFT markets, Easley et al. (2012) argue that traders operate on a volume basis and focus on turning over holdings within specific numbers of contracts traded, as opposed to over specific time intervals. We next explain the VPIN and BV-VPIN measures in detail and then analyze whether the latter's auxiliary information enables better prediction of adverse market conditions when applied to daily international index data.

3.1. Volume-Synchronized Probability of Informed Trading (VPIN)

VPIN is based upon the PIN model that Easley et al. (1996) put forward as a measure of information asymmetry between informed and uninformed trades. PIN is premised on uninformed traders buying or selling regardless of whether new information exists, and informed trading occurring only when new information exists and buying (selling) upon arrival of good (bad) news. PIN is not a directly observable measure and is a function of the theoretical parameters of a microstructure model that is calculated using Maximum Likelihood Estimation (MLE) by fitting a mixture of three Poisson distributions (Easley et al., 1997). However, the numerical optimization procedure often has difficulty converging and estimations may be biased (Easley et al., 2010; Lin and Ke, 2011; Abad and Yage, 2012). These difficulties are exacerbated with the sheer size of HFT data sets. PIN is based on a sequential trading model with Bayesian updates as shown in Equation (1).

$$PIN_t = \frac{\alpha_t \mu}{\alpha_t \mu + 2\epsilon} \quad (1)$$

The probability that new information will arrive within the time frame of the analysis is given by α , and μ is the arrival rate of informed traders, assumed to follow a Poisson process. The arrival

rate of uninformed traders is given by ϵ . Easley et al. (2008) propose a dynamic econometric micro-structure model of trading and show that for a sufficiently large μ , the expected trade imbalance ($E[|V^B - V^S|]$) and the expected total number of trades ($E[|V^B + V^S|]$) are given by Equations (2) and (3), respectively.

$$E[|V^B - V^S|] \approx \alpha\mu \quad (2)$$

$$E[|V^B + V^S|] \approx \alpha_t\mu + 2\epsilon \quad (3)$$

where V^B (V^S) is the volume traded against the Ask (Bid). Where PIN uses an itemized classification scheme to distinguish between buy and sell volume, Easley et al. (2012) propose an approach called “bulk classification” for VPIN. They argue that in a HFT environment, itemized approaches can be problematic and suggest that instead of a calendar clock, a volume clock is applied with a fixed Volume Bucket Size (VBS). Using VBS is advantageous as it is analogous to having a trading session split into periods of comparable information content and minimizes the impact of volatility clustering.

A trade classification algorithm is applied to identify the Buy (V^B) and Sell (V^S) volume. As all volume bars are of a fixed size, Easley et al. (2012) show that V can be given by Equation (4).

$$V = \frac{1}{N} \sum_{\tau=1}^N (V_{\tau}^S + V_{\tau}^B) = \alpha\mu + 2\epsilon \quad (4)$$

For a given number of N volume bars, based on Equations (1), (2), (3) and (4) it can be shown that VPIN is a good approximation of PIN and is given by Equation (5).

$$VPIN = \frac{\sum_{\tau=1}^N |V_{\tau}^S - V_{\tau}^B|}{N \times V} \quad (5)$$

A high VPIN value indicates large order imbalances that may threaten liquidity provisions, hence its application as a market liquidity indicator. With access to tick data or order book information, VPIN can be operationalized by adding the buy and sell volume of each volume bar. However, tick data and market order data are often inaccurate, incomplete, or expensive to access. Thus, trade classification algorithms are often used to approximate the buy or sell volume.

3.2. Bulk Volume VPIN (BV-VPIN)

VPIN is often applied with one of three trade classification algorithms: Tick Rule VPIN (TR-VPIN), BV-VPIN, and Lee-Ready VPIN (LR-VPIN). Both TR-VPIN and BV-VPIN are Level-1 algorithms that only use trade price data. LR-VPIN is a Level-2 algorithm that uses both trade and

quote data, and is commonly applied in markets where it is difficult to distinguish the initiating³ side of the trade.

Chakrabarty et al. (2012) report that the BV-VPIN is a more time-efficient methodology whereas TR-VPIN produces more accurate measures of order imbalance and order flow toxicity. They find that both the TR-VPIN and BV-VPIN perform well when identifying the aggressor side of trading, with the TR-VPIN being slightly more accurate. Easley et al. (2016) examines the accuracy and efficiency of the BV-VPIN, LR-VPIN and Aggregated Tick Rule. The authors conclude that tick rule approaches are relatively good classifiers of aggressor trades. However, BV-VPIN is shown to be sufficiently accurate for classifying buy and sell trades, and goes beyond tick-based approaches by providing insight into other proxies for underlying trade information. Moreover, it has the advantage of requiring substantially less data to implement.

In our study involving the analysis of daily data, we apply bulk volume classification to calculate the probability (i.e., portion) of buy and sell in each volume bar. Trades are aggregated over volume intervals and the price change between the two consecutive intervals is used to approximate the percentage of buy and sell order flow. The amount of volume that is classified as Buy ($\hat{V}_\tau^B$) and Sell ($\hat{V}_\tau^S$) is given by Equations (6) and (7), respectively.

$$\hat{V}_\tau^B = V_\tau \cdot t \left(\frac{P_\tau - P_{\tau-1}}{\sigma_{\delta p}}, df \right) \quad (6)$$

$$\hat{V}_\tau^S = V_\tau \cdot \left[1 - t \left(\frac{P_\tau - P_{\tau-1}}{\sigma_{\delta p}}, df \right) \right] \quad (7)$$

where P_τ is the single price ascribed to each current volume bar and $P_{\tau-1}$ is the price of previous volume bar. $\sigma_{\delta p}$ represents the volume-weighted standard deviation of price change over two consecutive volume bars. A volume bar is denoted as τ . A Student t distribution is used to approximate the price changes, and probability of buy and sell volume is calculated by CDF of the empirical price change distribution. The underlying trade is more likely to be buyer-initiated (seller-initiated) if the price change is positive (negative) and larger in magnitude, relative to the distribution of past price changes. Volume-weighted standard deviation is calculated by Equation (8).

$$\sigma_{\Delta P_i} = \sqrt{\frac{\sum_{\tau=1}^n V_{i,\tau} (\Delta P_{i,\tau} - \overline{\Delta P_i})^2}{\sum_{\tau=1}^n V_{i,\tau}}} \quad (8)$$

³Baker and Kiymaz (2013) state that order flow is defined as the number of trades where the buyer was the aggressor minus the number of trades where the seller was the aggressor. In a limit-order market (OTC market), the aggressor is the agent placing a market order (requesting a quote).

4. Results

Easley et al. (2016) show that choices of the VBS and sample length greatly impact the implementation and estimation of BV-VPIN, and that the suitable selection of these variables is dependent on optimizations being performed and should reflect the nature of trades within the market investigated. In Subsection 4.1 we discuss the choices available for different parameters of BV-VPIN and the values selected in our study across different markets.

To protect against the risks of adverse selection and preserve the integrity of the liquidity provision process, two approaches have been proposed. First, Zweig (2012) reports that exchanges can dynamically adjust the speed of the trading engine. If bids are being struck at such a speed that market makers are unable to adequately fill liquidity demands, they will be forced out resulting in a liquidity crash. Thus, the trading engine would decelerate (accelerate) the speed of matches occurring at the bid (ask) price. In such an approach BV-VPIN can be used as early-warning indicator for when exchanges should take action, and super-computing resources may be applied accordingly (Wu et al., 2013). By calculating BV-VPIN for multiple international markets, we analyze VPIN's ability to forecast sharp downward movements at different thresholds. We take the view of a global-macro fund manager who decides to invest in each country's equities index when risks are low, and switches to the risk-free rate when risks are high. Henceforth, the economic impact of VPIN is evaluated by switching to the risk-free rate (i.e., US 1-month T-bills) when VPIN levels are high, and investing in equities index when VPIN levels are low. We report the outcomes of this analysis in Subsection 4.2.

Second, Easley et al. (2011a) outline specifications of a VPIN futures contract, that offers market makers a hedge against the risk of higher than expected levels of toxicity (i.e., protection against the rise in the probability of adverse selection for liquidity providers). Market makers would purchase protection when their inventories are rising beyond normal levels. Informed traders would sell that protection once their orders have been fulfilled, thus allowing them to monetize their private knowledge that their contribution to toxicity has ceased. In Subsection 4.3, we analyze the safe haven and hedge properties of the VPIN if implemented as a futures contract in accordance with Easley et al. (2011a). Using the methodology of Baur and McDermott (2010) and Low et al. (2016), we analyze whether the VPIN of each country exhibits safe haven and hedging properties versus each country's equities index during the extreme quintiles of the equities returns distribution (e.g., 10%, 5%, 1%) and during well-known crisis periods.

4.1. Parameter selection for BV-VPIN

BV-VPIN takes price and volume as input parameters. Table 2 lists the key parameters to choose when calculating BV-VPIN. We investigate the optimal parameters based upon the US dataset and apply the same parameters for all other countries throughout our exploratory study. The US

Table 2: Parameters of BV-VPIN model

This table presents the key parameters of BV-VPIN model when applied in daily data. The values used in our analysis is given in the third column.

Description	Parameters	Parameters Selected
Number of volume bars	1000, 2000, 3000, 4000	2000
Rolling window size (bars)	10, 20, 50, 100	50
Distribution for price changes between volume bars	Normal, Student t (df=0.1, 0.25, 1, 5, 10)	Student t (df=5)
Student's t distribution with df=5	Normal, Log-normal	Log-normal

equities market continues to be one of the largest and most heavily traded markets globally, and therefore is an appropriate choice for a test case.

The bulk volume classification algorithm requires each volume bar to be associated with a single price. Wu et al. (2013) examine the performance under different prices methodologies (e.g., average, weighted average, median, and weighted median) and find that the results do not differ significantly. We follow Easley et al. (2016) who suggest using the last price of each volume bar to represent the price of the bar.

Prior literature applies fixed values (e.g., 1000, 5000) for each volume bar. Our study calculates BV-VPIN across equity indices from six countries where each market has different trading volumes. Therefore, it is impractical to use a single value for the volume bar for all different markets. Thus, we fix the total number of bars such that the volume of each bar is approximately equal to the total volume divided by the number of bars.



Figure 1: BV-VPIN and the CDF of BV-VPIN calculated for the US market (S&P500) with different volume bar sizes.

Figure 1 shows the BV-VPIN of the US market with different volume sizes, keeping all other parameters fixed. Small volume sizes produce a more volatile BV-VPIN, increasing the degree of noise into BV-VPIN's CDF. Applying larger volume sizes “smoothens” the BV-VPIN but may result in over-fitting the BV-VPIN such that large impending price movements in the underlying equities index go undetected. For each country investigated, the optimal choice for the volume bar size may differ. From Figure 1, a selection of 200 for the total number of volume bars looks to have roughly the right balance of capturing salient downward features in the S&P500 index.

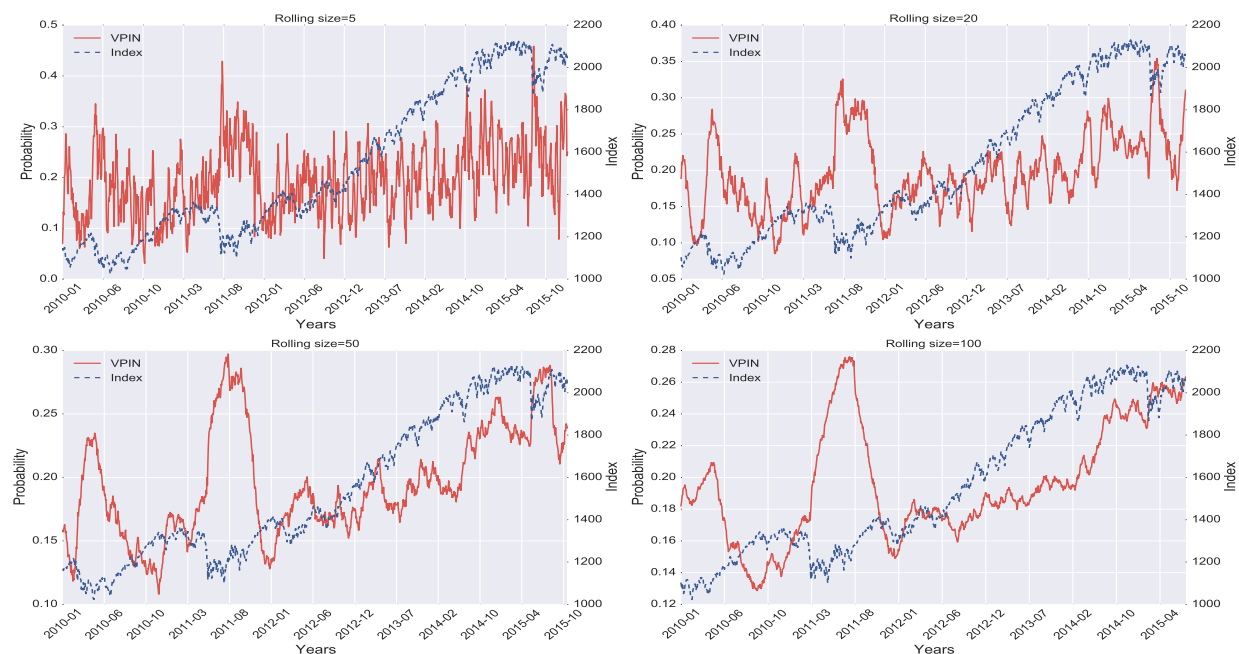


Figure 2: BV-VPIN and the CDF of BV-VPIN calculated upon the US market (S&P500) where rolling-window lengths vary.

The rolling window size (or volume buckets) is another important input parameter. Figure 2 shows the marginal effect on the BV-VPIN curve of different rolling window sizes. Smaller window sizes follow price movements more closely but exhibit higher volatility. Larger window sizes result in less volatile and smaller BV-VPIN peaks. As our application of BV-VPIN is in the context of a long-term indicator of market liquidity in a low-frequency environment, we select 50 as the window size to ensure that large adverse price movements are detected.

BV-VPIN classifies trade volume based on the distribution of price changes over volume bars. Easley et al. (2012) suggest that for high frequency data, a suitable choice is Student's t distribution with a df of 0.25. For daily data, we have fitted our distributions using both Normal and Student's t distributions. Figure 3a is the kernel density of price changes versus the Normal and Student's t distribution. It is intuitive to expect the Student's t distribution to provide a superior fit to that of the Normal distribution, due to the prevalence of kurtosis within our data set of international

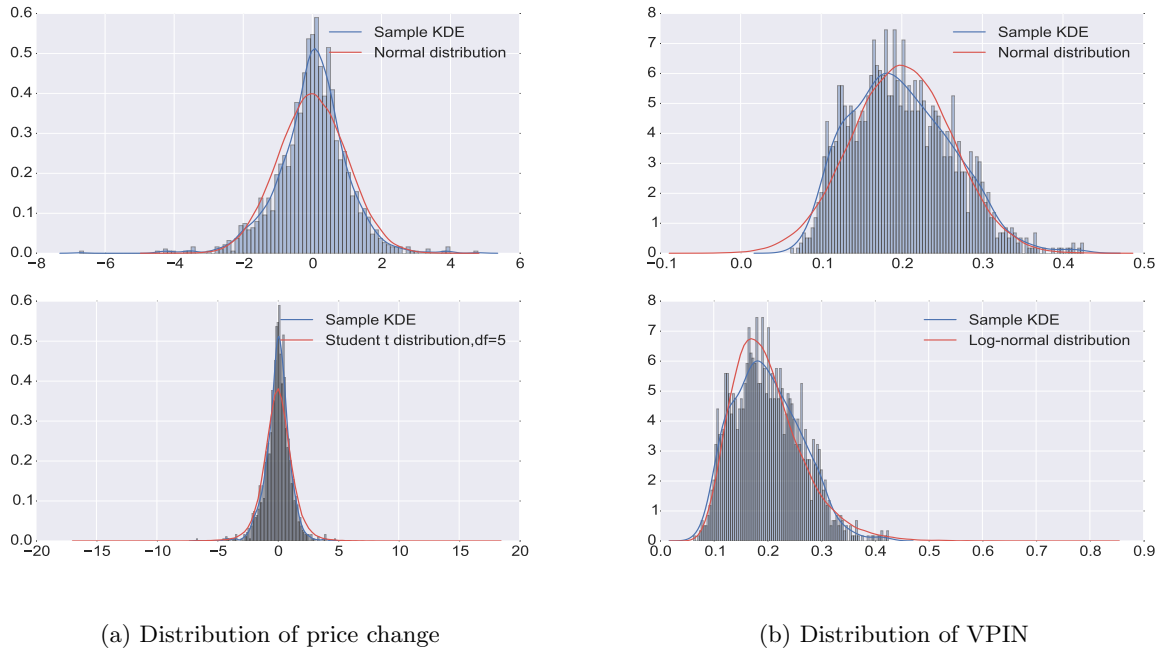


Figure 3: Distribution fitting of price changes (across volume bars) of BV-VPIN of the US market (S&P500).

equities indices. Upon applying both the Normal and Student's t distribution, our results show no obvious differences between the two. We select the Student's t distribution with $df=5$.

Applying BV-VPIN alone is not an effective early warning signal. Easley et al. (2011b) shows that the BV-VPIN rises almost simultaneously during prices drops and reaches its peak when prices have stabilized. Easley et al. (2014) shows that the CDF of BV-VPIN reaches its peak before BV-VPIN, and successfully predicts the impending "flash crash" in the E-mini S&P500. As the BV-VPIN is calculated using absolute volume balances, we apply the log-normal distribution to approximate the VPIN distribution. Figure 3b shows that the sample distribution of the S&P500 VPIN is more likely to follow a log-normal distribution. Wu et al. (2013) suggest that the log-normal distribution accurately describes the BV-VPIN sample. They also recommend using the truncated log-normal distribution to avoid the occurrence of extremely small BV-VPIN values. For daily data, as BV-VPIN values are significantly larger than zero, we use the ordinary log-normal distribution with the CDF given by Equation (9).

$$\frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{\ln x - \mu}{\sigma \sqrt{2}} \right) \right] \quad (9)$$

where erf is the Gaussian error function, μ is the average of the log of BV-VPIN values, and σ is the standard deviation.

4.2. BV-VPIN as an early warning signal

Figure 4 presents the BV-VPIN and CDF of BV-VPIN for equity indices in different countries. We define a “VPIN event” to occur when the CDF of BV-VPIN exceeds a threshold. Given that BV-VPIN follows a lognormal distribution, a threshold may be selected based on mean and variation⁴. In our implementation, we select 0.6, 0.8, and 0.9 as intuitive threshold values.

BV-VPIN tends to stabilize at 0.2 and then rises in concert with large price movements in the underlying index. For example, in Figure 4a, the all-time high occurs during 2008-2009, also the most volatile period for the S&P500. It is noticeable across all countries that BV-VPIN is negatively correlated with level movements in the underlying index. However, BV-VPIN increases almost simultaneously when the index starts to fall.

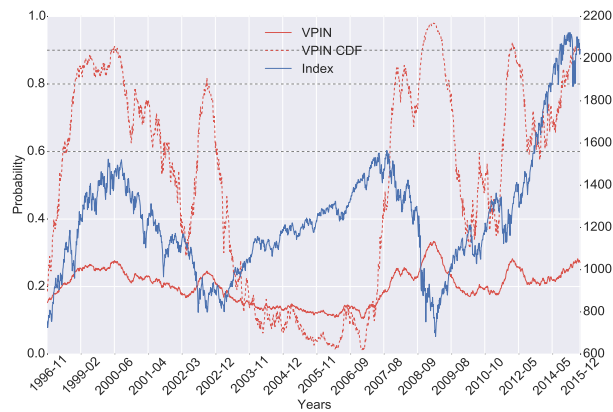
The CDF of BV-VPIN rises earlier and quicker than BV-VPIN and is more effective as an early-warning signal. All markets exhibit high BV-VPIN levels during well-known crises (e.g., 2008 Great Recession, 2011 US Credit Rating Downgrade). In the later months of our dataset, we can see that BV-VPIN is relatively high, indicating that markets remain susceptible to volatility and liquidity shocks despite the best efforts of financial regulators and central banks to improve credit risk and market risk exposures within the financial system.

As BV-VPIN is proportional to the absolute difference between buy and sell volume, any sudden and large upward price movements should result in BV-VPIN rising sharply. However, empirically, we are more likely to observe large and sudden price drops across US and international equities (Longin and Solnik, 1995; Ang and Chen, 2002; Low et al., 2013). Thus, our analysis of prediction accuracy investigates the effectiveness of CDF of BV-VPIN in forecasting impending index falls.

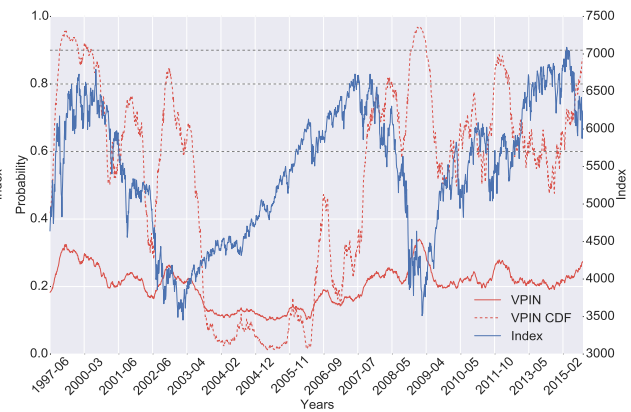
Table 3 shows the prediction accuracy with different thresholds applied. With 0.9 as a threshold value, we have 100% prediction accuracy in 4 markets including US, UK, China and Australia, and 75% accuracy in Germany. Using the same threshold, only one BV-VPIN event is observed in Japan. With a threshold value of 0.8, we have 75% prediction accuracy in only two countries. Moreover, prediction accuracy is less than 50% in all countries if using 0.6 as threshold. It is intuitive to expect that lower thresholds will increase the number of false alarms, whereas a higher threshold might miss market downturns. The results presented in Table 3 are based on Figure 4. As BV-VPIN can hover around the threshold values, we only consider a BV-VPIN event to be the first breach of the threshold value.

Intuitively, when a BV-VPIN event occurs, it is important to evaluate both the magnitude of the fall in the underlying index, and the time horizon over which it occurs. This allows the portfolio manager to understand how much time he has to exit all of his equity position if he decides to heed the BV-VPIN signal. Table 4 presents both the prediction accuracy and the index downturn with respect to different time horizons. We report our results applying a threshold value of 0.9 for all

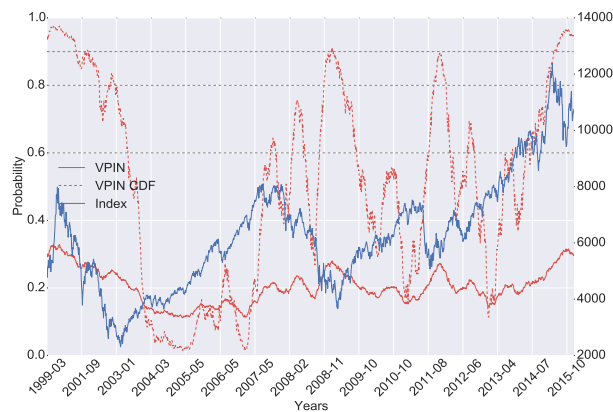
⁴Wu et al. (2013) use two standard deviations greater than the average value to denote a “VPIN Event”.



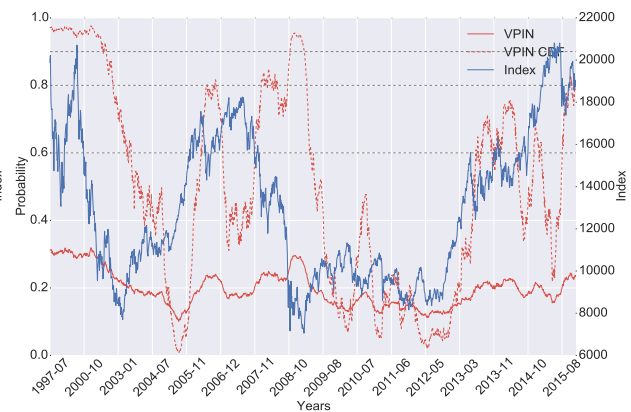
(a) United States



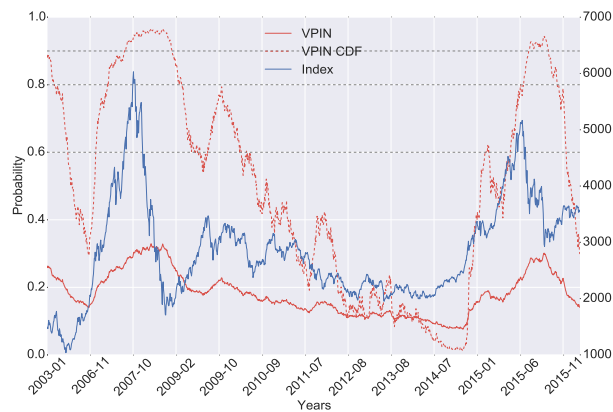
(b) United Kingdom



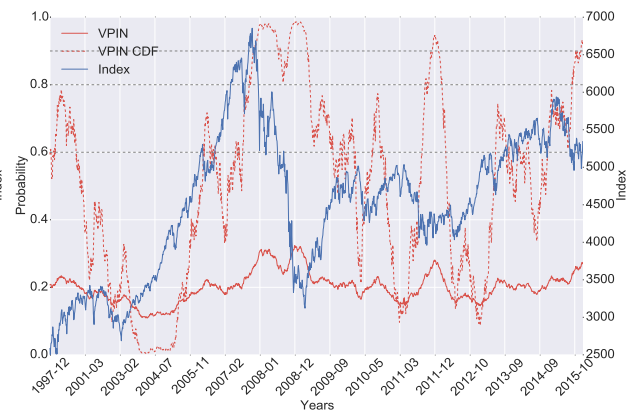
(c) Germany



(d) Japan



(e) China



(f) Australia

Figure 4: BV-VPIN and CDF of BV-VPIN for international equities indices.

Table 3: Prediction Properties of BV-VPIN

This table presents the prediction properties of BV-VPIN; a BV-VPIN Event occurs when the CDF of BV-VPIN exceeds the threshold value. Price drops are ascertained by any negative return that occurs within a horizon of 30 to 60 days after the BV-VPIN event occurs. Prediction accuracy is determined as the number of actual negative returns on the respective index, divided by the total number of BV-VPIN events.

Threshold	Number of BV-VPIN Events	Number of Price drops	Prediction Accuracy
<i>Panel A: United States</i>			
0.9	4	4	100%
0.8	5	3	60%
0.6	5	2	40%
<i>Panel B: United Kingdom</i>			
0.9	3	3	100%
0.8	4	3	75%
0.6	14	6	43%
<i>Panel C: Germany</i>			
0.9	4	3	75%
0.8	5	3	60%
0.6	8	3	38%
<i>Panel D: Japan</i>			
0.9	1	0	0%
0.8	4	3	75%
0.6	10	3	30%
<i>Panel E: China</i>			
0.9	2	2	100%
0.8	1	2	50%
0.6	3	1	33%
<i>Panel F: Australia</i>			
0.9	3	3	100%
0.8	4	1	25%
0.6	7	1	14%

Table 4: Prediction Horizon of BV-VPIN

This table presents the prediction properties of BV-VPIN when using fixed thresholds of 0.9 for all countries, except Japan. For Japan, a threshold value of 0.8 is used. Prediction accuracy denotes the number of index drops out of the total number of BV-VPIN events where the CDF of BV-VPIN exceeds the threshold value. The Average Index Change indicates the average index returns based upon the number of days after a BV-VPIN event occurs. The Average Index Change (Negative) indicates the average index returns only when the BV-VPIN event precedes a drop in the index.

Prediction Horizon (Days)	Prediction Accuracy	Average Price Drop (Negative Returns)	Average Price Drop (All Returns)
<i>Panel A: United States</i>			
5	3/4	-3.83%	-1.09%
20	3/4	-7.00%	-3.27%
60	2/4	-5.42%	2.74%
120	2/3	-13.06%	-4.60%
<i>Panel B: United Kingdom</i>			
5	2/3	-3.04%	4.03%
20	2/3	-6.81%	-2.46%
60	1/3	-18.41%	-4.72%
120	2/3	-2.70%	-1.25%
<i>Panel C: Germany</i>			
5	1/3	-8.55%	-0.16%
20	1/3	-1.12%	2.15%
60	2/3	-4.43%	-2.26%
120	3/3	-3.33%	-3.33%
<i>Panel D: Japan</i>			
5	3/4	-4.33%	-3.05%
20	3/4	-4.87%	-2.10%
60	2/3	-9.44%	-5.82%
120	2/3	-9.48%	-5.86%
<i>Panel E: China</i>			
5	0/2	n.a.	1.62%
20	1/2	-4.05%	4.88%
60	1/2	-19.15%	-0.06%
120	2/2	-5.57%	-5.57%
<i>Panel F: Australia</i>			
5	3/4	-3.60%	-2.44%
20	3/4	-2.35%	-1.51%
60	2/3	-16.04%	-10.67%
120	2/3	-20.20%	-13.26%

countries except Japan, for which a threshold value of 0.8 is used. We record the dates associated with each BV-VPIN Event, and use the index level at these dates as the benchmark level. We observe prices 5, 20, 60 and 120 trading days after the BV-VPIN Event. Based on these values, we calculate the fall in the index for each BV-VPIN Event over different time horizons.

The calculation for the average index fall for negative returns is based solely upon BV-VPIN events that resulted in an index fall. For example, over the one-month prediction horizon for the US index, there are four BV-VPIN events, two of which resulted in index falls and two that were false alarms. The calculation for the average price drop for all returns is based upon all BV-VPIN events regardless of whether there was a index drop after the event. This allows us to understand the average index drop if BV-VPIN thresholds are breached whilst taking into account the impact of false alarms. The average index drop after a BV-VPIN event is calculated as given by Equation (10).

$$PD_{avg,t_f} = \frac{1}{t_f} \sum_{i=1}^{t_f} R_i \quad (10)$$

$$R_i = \frac{P_i - P_{t_0}}{P_{t_0}} \quad (11)$$

The day of the BV-VPIN event occurs is given by t_0 , and t_f represents 5, 20, 60 or 120 trading days after the BV-VPIN event. When calculating the average price drop for negative returns, we only include $R_i < 0$. In this scenario, the average price drop is solely based on the average of the two BV-VPIN events that resulted in price drops. In this manner, we understand the magnitude of the price drop when the BV-VPIN threshold is breached and a price drop has been accurately predicted.

Table 4 shows that prediction accuracy and the average price drops vary across different markets and prediction horizons. We see that when the BV-VPIN event accurately predicts a price drop, we can expect to see a negative return of more than 9% in the following period of 60 or 120 days (except for Germany). However, within the first 5 days, we observe price drops of about 5% or less (except for China and Germany). One can conclude that once the BV-VPIN threshold is breached, a portfolio manager has at least 30 days to exit his positions to minimize the magnitude of losses. When accounting for all BV-VPIN events (i.e., including the impact of false alarms), we see that after 120 trading days, all six countries experienced drops in the index ranging from -1.25% (UK) to -13.26% (Australia). Within the first 5 and 20 trading days, we observe an index drop in 4 countries. Our results indicate that there are substantial drops in each country's index after a BV-VPIN event.

4.3. BV-VPIN as a hedge and safe haven investment

To evaluate the hedge and safe haven properties of BV-VPIN if implemented as a futures contract, our analysis utilizes the following model as shown in Equation (12)-(14).

$$r_{VPIN,t} = a + b_t r_{index,t} + \varepsilon_t \quad (12)$$

$$b_t = c_0 + c_1 D(r_{index} q_{10}) + c_2 D(r_{index} q_5) + c_3 D(r_{index} q_1) \quad (13)$$

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1} \quad (14)$$

Equation (12) models the relationship between the CDF of BV-VPIN ($r_{VPIN,t}$) and each individual country's equity index ($r_{index,t}$). The parameter b_t is modeled as a dynamic process given by Equation (13). Indicator variables ($D(\dots)$) are applied to capture extreme market movements, taking a value of one if stock market return at time t exceeds a certain threshold given by the 10%, 5% and 1% quantile of the return distribution, and zero otherwise. The residual term ε_t is modeled as a GJR-GARCH process (Glosten et al., 1993) that takes into account volatility clustering and asymmetries in the ARCH process. All equations are jointly estimated with MLE. Such models have been applied in analyzing the safe haven and hedge properties of commodities such as precious metals (Baur and McDermott, 2010) and diamonds (Low et al., 2016).

Table 5: Hedge and safe haven properties of BV-VPIN

This table presents the estimation results for the coefficients in Equation 13. Negative coefficients in the hedge column (c_0) signify that the asset is a hedge against index. Zero (negative) coefficients in extreme market conditions, namely quantile 10% (c_1), 5% (c_2), and 1% (c_3) indicate that the asset is a weak (strong) safe haven. Each column of t-statistics is associated with the coefficient column to the left, as an indication of the significance level.

Countries	Indices	Hedge	t-stats	Safe haven quantiles					
				10%	t-stats	5%	t-stats	1%	t-stats
US	S&P 500	0.055	3.373	-0.178***	-3.846	0.033	0.697	-0.041	-1.187
UK	FTSE 100	0.060	3.566	-0.191***	-4.160	0.104	2.263	-0.144***	-3.763
Germany	DAX	0.051	4.248	-0.155***	-4.188	0.012	0.319	-0.061**	-2.003
Japan	Nikkei 225	0.041	2.971	-0.133***	-3.930	-0.003	-0.010	-0.062**	-2.261
China	SHCOMP	0.144	8.356	-0.286***	-5.542	-0.021	-0.382	-0.017	-0.408
Australia	AS30	0.090	3.690	-0.207***	-3.456	-0.130**	-1.967	-0.018	-0.341

*, **, *** indicates statistical significance at the 10%, 5%, and 1% level, respectively.

If any of the parameters (c_1 , c_2 , c_3) are significantly different from zero, this suggests a relationship between BV-VPIN and the country's index. If the parameters in Equation (13) are non-positive, BV-VPIN acts as a weak safe haven. If the parameters are negative and statistically different from zero, BV-VPIN becomes a strong safe haven. Where c_0 is zero or negative, and the sum of the parameters c_1 to c_3 are not jointly positive (exceeding the value of c_0), BV-VPIN serves as a hedge, where negative c_0 suggests strong hedging attributes and a value of zero indicates weak hedging attributes.

The estimated results of the regression model given by Equations (12), (13), and (14) are reported in Table 5 which contains the estimates of c_0 , c_1 , c_2 and c_3 . Under the GJR-GARCH model, it is shown that BV-VPIN is a strong safe haven in all six countries within the 10% quantile with strong statistical significance. At the 1% quantile, BV-VPIN remains an effective safe haven in all six countries, particularly in UK, Germany, and Japan where the coefficients are statistically

significant. Only Japan, China, and Australia exhibit safe haven qualities at the 5% level, with Australia's results being statistically significant. All six countries have positive coefficients for c_0 , suggesting that BV-VPIN does not exhibit suitable qualities for hedging.

Table 6: Hedge and safe haven properties of BV-VPIN during periods of financial stress

This table presents the estimation results for the coefficients in Equation 15. The duration of crisis periods is set to be 20 days after the crisis starts. Negative coefficients in the hedge column (c_0) signify that BV-VPIN is a hedge against the market. Negative coefficients in subsequent columns show that it is a safe haven during the AFC (c_1), DCB (c_2) GR (c_3) or USCRD (c_4). Each t-statistics column is associated with the coefficient column to the left, as an indication of the significance level.

Countries	Hedge		1997 AFC		2000 DCB		2008 GR		2011 USCRD	
	Coeff.	t-stats	Ttl. eff.	t-stats	Ttl. eff.	t-stats	Ttl. eff.	t-stats	Ttl. eff.	t-stats
US	-0.004	-0.365	0.022	0.123	-0.055	-0.202	-0.150***	-2.798	-0.192***	-2.861
UK	0.004	0.279	0.353	2.014	-0.170	-0.570	0.503	16.880	-0.302***	-2.841
Germany	-0.010	-1.173	n.a.	n.a.	-0.020	-0.106	-0.408***	-6.907	-0.100**	-2.133
Japan	-0.019**	-2.139	-0.087	-1.008	-0.053	-0.198	-0.056	-0.766	0.221	2.620
China	0.003	0.227	n.a.	n.a.	n.a.	n.a.	0.088	0.782	-0.224**	-2.493
Australia	-0.035**	-2.118	n.a.	n.a.	0.236	0.759	-0.083	-0.983	-0.140	-1.147

*, **, *** indicates statistical significance at the 10, 5, and 1 percent level, respectively.

Equation 15 allows the analysis of hedge and safe haven properties during crisis sub-sample periods. By identifying well known financial crisis periods, we apply time dummies equal to one if the returns fall within the pre-defined period and a zero otherwise. We define starting dates and assume that most of the crisis and its effects occurred in the first 20 trading days (approximately one month) from the start date (Baur and McDermott, 2010). Four major financial events occurred in the sample period of our investigation, namely the Asian Financial Crisis (AFC) (Oct. 22, 1997 - Nov. 19, 1997), the Dot-com Bubble (DCB) (Mar. 10, 2000 - Apr. 7, 2000), the Great Recession (GR) (Sept. 12, 2008 - Oct. 2, 2008) and the US Credit Rating Downgrade (USCRD) (Jul. 23, 2011 - Aug. 12, 2011).

$$b_t = c_0 + c_1 D(\text{AFC}) + c_2 D(\text{DCB}) + c_3 D(\text{GR}) + c_4 D(\text{USCRD}) \quad (15)$$

If the parameters c_1 , c_2 , c_3 or c_4 are zero or negative, BV-VPIN is a safe haven during the respective crisis period. Alternatively, a positive parameter means that the asset moves in tandem with the market and is not a safe haven.

Table 6 presents the estimation results of the model in Equation (15). For Germany, China, and Australia the results for AFC and DCB are unavailable due to the sample length of our dataset and the structure of our empirical analysis applied to calculate BV-VPIN. When calculating BV-VPIN, we group trade data across several days into one volume bar, then compute BV-VPIN based on a rolling window of 50 volume bars. Thus, there are no BV-VPIN values for the initial 49 volume bars. Moreover, the trading volume is relatively small in the earlier time periods of our data set. If a fixed-size volume bar is applied, each of the initial volume bars contains 5-10 days of data.

Thus, for certain countries the BV-VPIN values are unavailable for the initial months or years in our data set.

We find that three countries have BV-VPIN values associated with the AFC, and only in Japan does BV-VPIN exhibit some safe haven properties. During the DCB, all countries except Australia exhibit safe haven properties. During the GR, BV-VPIN exhibits safe haven qualities in four countries, particularly in the US and Germany where the coefficients are statistically significant at 1%. In the UK and China, BV-VPIN is not an effective safe haven during the GR. BV-VPIN is a strong safe haven in all countries except for Japan during the period around the US credit downgrade. It exhibits statistical significance in US, UK, Germany, and Australia. In four countries out of six, BV-VPIN exhibits hedging qualities. In both Japan and Australia, significant negative coefficients are observed, thus indicating strong hedge properties against market crises. These results show that if the BV-VPIN is operationalized as a futures contract, it would exhibit suitable safe haven and hedge characteristics for asset managers who wish to protect themselves from the high-volatility and liquidity shocks that may occur from order imbalances.

4.4. BV-VPIN in a risk-on/risk-off trading strategy

We test a risk-on/risk-off trading strategy based on the CDF of BV-VPIN. Investments are in equities (risk-on) when the CDF of BV-VPIN is below a set threshold, otherwise we invest in the risk-free rate (risk-off). Equities are represented by each country's stock index, and the risk-free rate used is the US 10-year treasury rate. We refer to this strategy as the BV-VPIN Strategy. For comparable benchmark strategies, we report the realized returns on a Buy-and-Hold investment in (1) Equities and (2) Risk-Free Rate (RF).

Table 7 represents the performance of these three strategies for each of six different countries. Similar to the analysis of prediction accuracy in Subsection 4.2, threshold values of 0.6, 0.8, and 0.9 are applied. We report descriptive statistics such as the mean, standard deviation, skewness, and kurtosis. Risk-adjusted returns performance is given by the Sharpe Ratio (Sharpe, 1994) and the Sortino Ratio (Sortino and Satchell, 2001) that capture total volatility and downside volatility, respectively. The initial investment for each approach is assumed to be \$100, and we report the Final Value (FV) and the Total Return (TR) that is reported in dollars and percentages, respectively.

Across all countries with the exception of Germany, all the strategies exhibit negative skewness, the highest being the BV-VPIN strategies at a threshold of 0.6 and the lowest being the RF strategy. Aside from RF, across all countries the BV-VPIN and Equities strategies exhibit excess kurtosis.

For all six countries, investment in the RF produces the highest Sharpe Ratio due to its low values of standard deviation. The BV-VPIN strategy with the optimal threshold outperforms the Equities strategy. In the US market, the BV-VPIN strategy with a threshold of 0.9 has the highest Sharpe ratio. In China, the Sharpe ratio of using BV-VPIN strategy is double that of the Equities strategy. In Japan, BV-VPIN strategy with a threshold of 0.6 produces a Sharpe Ratio five times larger than the Equities strategy. As measured by the Sortino ratio, the BV-VPIN strategy with

Table 7: Evaluation of BV-VPIN in trading strategy

This table compares the performance of investing in (1) BV-VPIN; a risk-on/risk-off strategy that invests in the respective country's equities index (risk-free rate) when the BV-VPIN is below (above) a set threshold, (2) Equities; a Buy-and-Hold Strategy in each country's stock index, and (3) Risk-Free; a Buy-and-Hold Strategy in the form of US 10-year treasury rate. The distribution characteristics of each (e.g., Mean, Standard Deviation, Skewness, and Kurtosis) for each strategy are reported. Risk-adjusted returns are reported with the Sharpe and Sortino Ratios. The final value of investment by investing an initial sum of \$100 is given by FV (\$), and TR (%) is the total return of the strategy. The Sortino Ratio for the risk-free rate is not applicable as the RF rate is never negative in our dataset.

Strategy	Threshold	Mean	Std. Dev.	Skew	Kurt.	Sharpe	Sortino	FV (\$)	TR(%)
<i>Panel A: United States</i>									
BV-VPIN	0.9	0.026	1.123	-0.158	7.363	0.024	0.033	263	163.17
	0.8	0.020	0.941	-0.205	5.907	0.021	0.029	207	107.41
	0.6	0.015	0.661	-0.435	10.382	0.023	0.032	187	86.51
Equities		0.029	1.252	-0.043	7.616	0.023	0.033	273	172.97
Risk-Free		0.011	0.004	-0.025	-0.934	2.974	n.a.	169	69.11
<i>Panel B: United Kingdom</i>									
BV-VPIN	0.9	0.012	1.099	-0.132	6.752	0.011	0.016	135	34.82
	0.8	0.007	0.992	-0.343	7.571	0.007	0.010	112	11.71
	0.6	0.013	0.619	-0.314	13.662	0.022	0.030	172	71.87
Equities		0.014	1.224	-0.009	5.491	0.011	0.016	134	34.02
Risk-Free		0.011	0.004	-0.069	-0.979	3.030	n.a.	165	65.34
<i>Panel C: Germany</i>									
BV-VPIN	0.9	0.028	1.407	0.152	6.867	0.020	0.029	218	118.50
	0.8	0.034	1.195	0.098	9.777	0.028	0.040	310	209.68
	0.6	0.021	0.869	-0.281	15.859	0.024	0.034	210	109.68
Equities		0.030	1.535	0.115	4.432	0.019	0.027	214	113.59
Risk-Free		0.010	0.003	0.036	-0.874	3.024	n.a.	155	55.22
<i>Panel D: Japan</i>									
BV-VPIN	0.9	0.029	1.243	-0.409	11.415	0.023	0.032	262	162.24
	0.8	0.021	1.100	-0.596	6.597	0.019	0.026	193	92.55
	0.6	0.035	0.891	-0.750	11.036	0.039	0.055	407	307.24
Equities		0.011	1.553	-0.146	5.440	0.007	0.010	96	-3.65
Risk-Free		0.011	0.004	-0.085	-0.976	3.044	n.a.	163	62.50
<i>Panel E: China</i>									
BV-VPIN	0.9	0.075	1.379	-0.364	4.091	0.054	0.078	773	673.27
	0.8	0.062	1.218	-0.224	5.211	0.051	0.074	562	461.98
	0.6	0.053	0.930	-0.239	7.927	0.057	0.084	463	363.61
Equities		0.044	1.686	-0.287	3.716	0.026	0.036	255	154.90
Risk-Free		0.009	0.003	-0.100	-1.258	3.306	n.a.	133	33.14
<i>Panel F: Australia</i>									
BV-VPIN	0.9	0.031	0.829	-0.251	3.003	0.037	0.052	346	246.05
	0.8	0.031	0.772	-0.346	3.168	0.040	0.057	361	260.79
	0.6	0.017	0.577	-0.583	6.035	0.029	0.040	198	98.17
BH		0.021	0.967	-0.431	5.317	0.022	0.030	209	108.99

the optimal threshold also outperforms Equities in all countries with the exception of the US. In the US market, BV-VPIN strategy has the same Sortino ratio as Equities. Generally, we find that the BV-VPIN strategy results in higher risk-adjusted returns.

When analyzing the final wealth value and total return of each strategy, application of the optimal BV-VPIN threshold outperforms both the Equities and RF with the exception of the US. With a threshold of 0.9, the final wealth of the BV-VPIN strategy comes close to the Equities for the US. For the UK, equities generally performed poorly in comparison to investing in RF. However, with a threshold of 0.6, the BV-VPIN strategy outperforms the both Equities and RF. In Germany, BV-VPIN strategy at thresholds of 0.9 and 0.8 successfully outperform both the Equities and RF strategies. In the Japanese market, Equities performs poorly compared to RF. However, applying the BV-VPIN strategy results in 307% returns with a threshold of 0.6, that is five times greater than the return from investing in the RF. The BV-VPIN strategy, regardless of threshold value, successfully outperforms the Equities and RF in China. With a threshold value of 0.9, the BV-VPIN strategy generates a fourfold return compared to the Equities, a threefold return for threshold of 0.8, and twofold return for threshold of 0.6. In Australia, application of the BV-VPIN strategy doubles the return on investment compared to the Equities strategy.

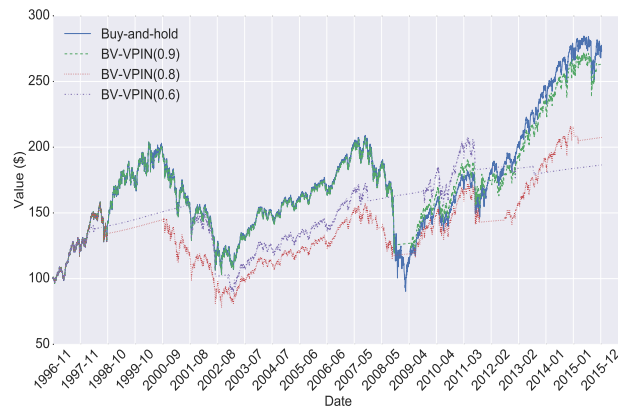
Figure 5 shows the accumulation of wealth with an initial investment of \$100 dollars using Equities and BV-VPIN strategies with different thresholds. We find that the BV-VPIN strategy effectively avoids large losses during extreme market downturns. For China and Australia, the BV-VPIN strategy suggests investing in the risk-free asset before the market drops and reinvesting in equity as market starts to recover. Specifically, we observe that in China, Australia, UK, and Japan the BV-VPIN avoids investing in equities during the GR period.

The optimal threshold to be applied for the BV-VPIN varies across different markets. For China and US, 0.9 is optimal, whilst in Australia and Germany, 0.8 produces the highest returns. In UK and Japan, 0.6 significantly outperforms higher thresholds. Our results indicate that one should tailor the choice of the optimal threshold depending on the market. However, application of a threshold value of 0.8 seems result in robust and profitable performance across all markets investigated in our study.

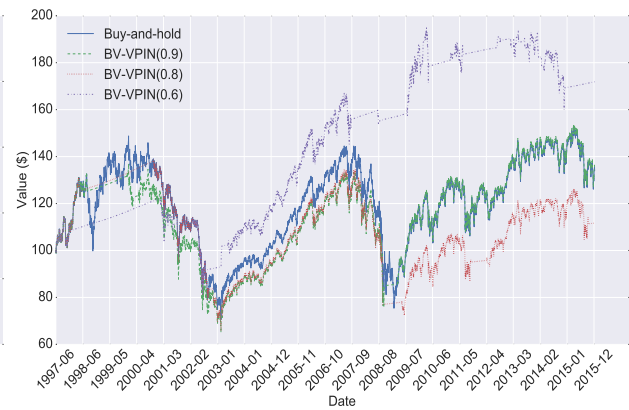
5. Conclusion

A structural model for market micro-structure research proposed by Easley et al. (1996) provides for the estimation of Probability of Informed Trading (PIN). The importance of PIN is that it provides a measure of order flow toxicity which can have a negative impact on market liquidity. With the advent of High Frequency Trading (HFT), Easley et al. (2012) develop a modified measure, Bulk Volume VPIN (BV-VPIN) where the probability of informed trading is based on volume imbalance and trade intensity.

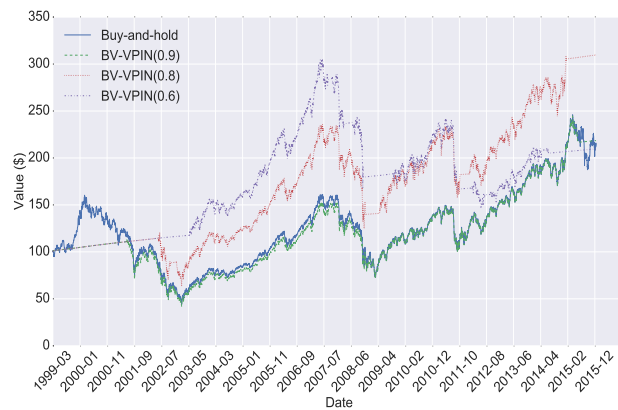
Market makers' estimates of time-varying toxicity level are a crucial factor in determining their



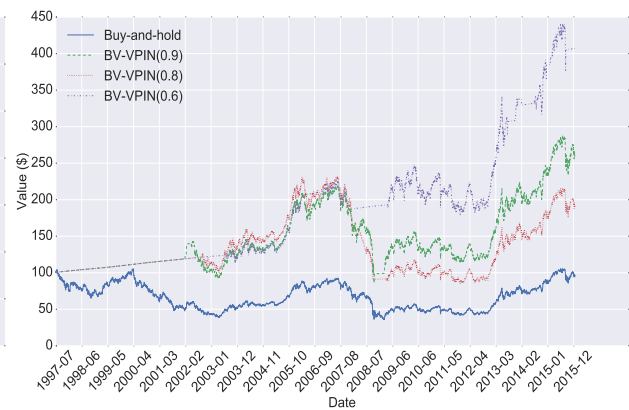
(a) United States



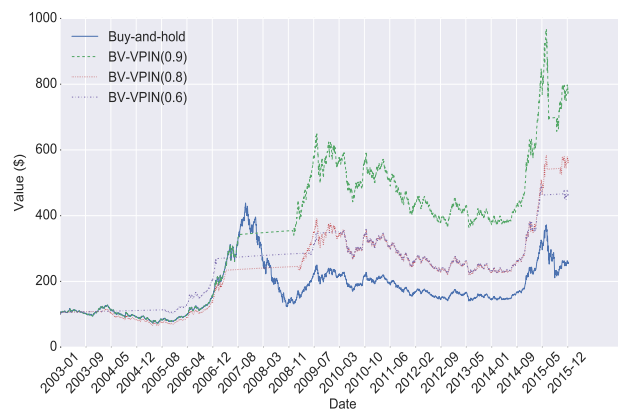
(b) United Kingdom



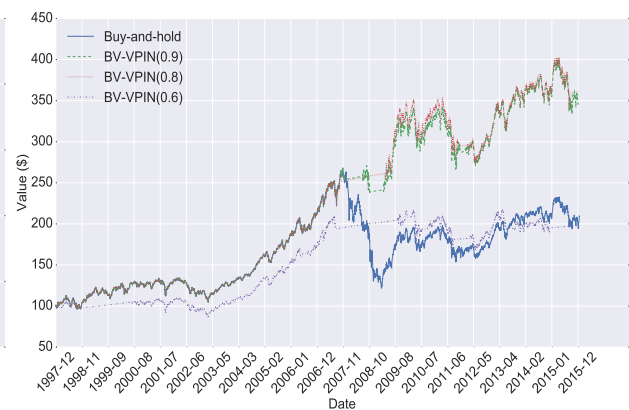
(c) Germany



(d) Japan



(e) China



(f) Australia

Figure 5: Accumulation of wealth of using different trading strategy for international equities indices.

participation such that if they believe that toxicity is high, they will liquidate their positions and leave the market, thus affecting market liquidity. Henceforth, BV-VPIN is a procedure that estimates the probability of informed trading based on volume imbalance and trade intensity.

Empirical tests suggest that BV-VPIN functions not only as an indicator of current market liquidity but also as an early warning signal for impending volatility. Although BV-VPIN itself is not effective in predicting future price movements, the cumulative distribution function (CDF) of BV-VPIN exhibits predictive properties in a HFT environment (Easley et al., 2011b). Abad and Yage (2012) propose that alternative specifications of the BV-VPIN model can result in alternative gauges for adverse selection. Thus, they suggest that BV-VPIN can be a broad measure of adverse selection beyond that of HFT applications.

We apply BV-VPIN on daily equities data across a range of international equities indices. We select optimal parameters for BV-VPIN based on the US market and apply it across a range of major global economies to ascertain its capabilities as a long-term indicator of market liquidity and impending market downturns. Our work includes the analysis of BV-VPIN's role as a safe haven and hedge instrument if implemented as a futures contract as suggested by Easley et al. (2011a). To investigate the economic significance of BV-VPIN as an early warning signal, we apply it in a risk-on/risk-off trading strategy where an asset manager invests in equities (risk-free rate) when the CDF of BV-VPIN is below (above) a threshold value.

We find that CDF of BV-VPIN exhibits predictive properties, however, the selection of threshold value can vary across different countries. This is in line with Easley et al. (2014) who contend that the selection of parameters of BV-VPIN needs to cater for the characteristics of the instrument being investigated. Our investigation of the hedge and safe haven properties of BV-VPIN suggests that it is an effective safe haven for most countries during extreme market downturns. BV-VPIN exhibits strong safe haven characteristics for most markets during three major crisis periods, the 2000 Dot-com Bubble (DCB), the 2008 Great Recession (GR) and the 2011 US Credit Rating Downgrade (USCRD). Incorporating BV-VPIN as part of a flight-to-quality trading strategy results in an outperformance of the Buy-and-Hold equities strategy for four out of six countries in our study. Specifically, in China, Japan and Australia, the BV-VPIN Strategy doubles the returns of a comparable buy-and-hold strategy in equities.

In conclusion, we show that BV-VPIN has applications beyond that of HFT environments. BV-VPIN is an effective indicator of long-term market volatility and exhibits hedging and safe haven characteristics during market downturns. BV-VPIN may also be used to improve portfolio management strategies for asset managers in ascertaining market conditions where one should minimize risk by reducing exposure to equities. For future work, other applications of BV-VPIN can involve investigating its ability to track the smart money effect as documented by in the managed mutual funds industry Gruber (1996); Zheng (1999); Sapp and Tiwari (2004) and tracking money flows between asset classes such as the flight-to-quality effect (equity to debt) by mutual fund

managers.

6. References

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